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# Large language models

*Weekend 1: Language models with the Ge'ez games*

Carlos Cotrini
*AI in Industry, ETH Zurich*
*Friday 16 October 2026* ፩፻፳፮

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Welcome, everyone

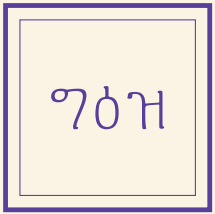
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# Large language models

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We start our first module with an understanding of large language models




*Introduction*

# Ge’ez

ግዕዝ

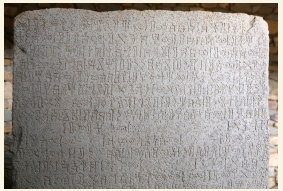
*Introduction*

# Ge’ez

To make things more vivid, I’m going to use the language “Ge’ez” as our illustrative example

# ግእዝ ቋንቋ::

- ጥንታዊ ደቡባዊ ሴማዊ ቋንቋ:: ይህ ቋንቋ ከአቢሲኒያ የመጣ ነው፤ አቢሲኒያ ዛሬ ኢትዮጵያና ኤርትራ ተብሎ ታወቃለች::
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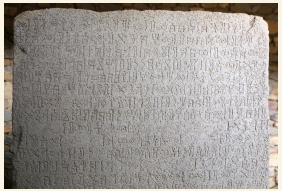





What is Ge’ez? Ge’ez is an ancient South Semitic language from the Horn of Africa

# ግእዝ ቋንቋ::

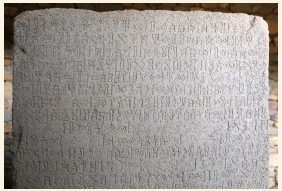
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originating in historical Abyssinia, today Ethiopia and Eritrea.

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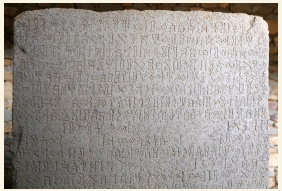





It is no longer a community's everyday spoken language

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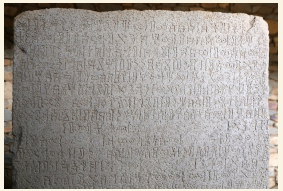




but it lives on as a liturgical language in several churches of the region

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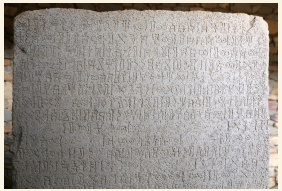





Ge’ez uses its own writing system, an abugida often called the “fidel,”

# ግእዝ ቋንቋ::

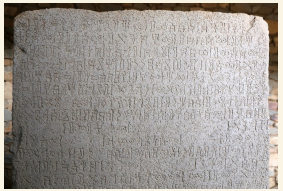
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where each sign represents a consonant with a built-in vowel

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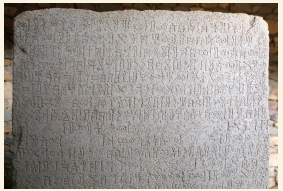
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Classical texts in Ge’ez include religious literature

# ግእዝ ቋንቋ::

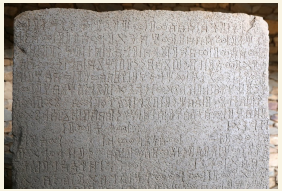
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royal chronicles, and translations of major works

# ግእዝ ቋንቋ::

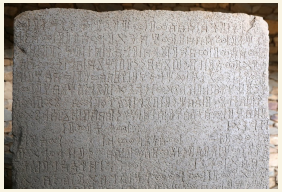
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giving us a window into the history of the region

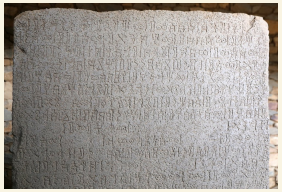
# The Ge’ez language

- Ancient South Semitic language. The language originates from Abyssinia, what is now Ethiopia and Eritrea.



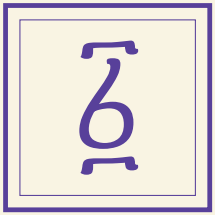
# The Ge’ez language

- Ancient South Semitic language. The language originates from Abyssinia, what is now Ethiopia and Eritrea.
- We use this language as basis to study and investigate modern large language models.



# The five Ge’ez games

| | | |
|---|---|---|
|  | ሥርዓት<br>**Structure** | We can extract structure from unfamiliar text without knowing a single word. |
|  | ምሳል<br>**Imitation** | By spotting patterns, we can compose new sentences in a language we don’t understand. |
|  | ትንቢት<br>**Prediction** | A language model picks the next word by probabilities conditioned on what came before. |
|  | ምልክት<br>**Representation** | A prefix becomes an embedding, and the embedding gives the chances of the next word. |
|  | ቅንብር<br>**Transformer** | An LLM computes the chances, draws a word, appends it, and repeats. |



*Section 1*

# Structure

# Game 1: objective

- We will show you some sentences in Ge'ez. They are related to either **city entities (city)** or **rural entities (countryside)**.

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# Game 1: objective

- We will show you some sentences in Ge'ez. They are related to either **city entities (city)** or **rural entities (countryside)**.
- Your task is to learn to discover when a sentence is talking about something in the city and when it is talking about something in the countryside.

| Countryside | City |
|---|---|
| *ʿEmäk lay ʾaḥar may* | *bet nawīd lay ʾaḥar ʿIqlēsīya* |
| *gāzā lay ʾasmāl may* | *maytān lay ʾasmāl ʿIqlēsīya* |
| *may lay ʾaḥar gāzā* | *ʿIqlēsīya lay ʾaḥar maytān* |
| *ṣadäq lay ʾaḥar ʿEmäk* | *bet masḥāf lay ʾaḥar bet nawīd* |
| *ṣadäq lay ʾaḥar gāzā* | *bet masḥāf lay ʾaḥar maytān* |

# Insight

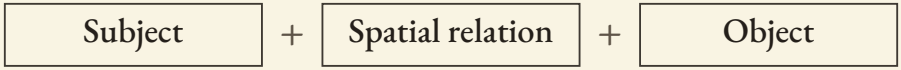
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# Insight

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- We are dealing with a very simple language:

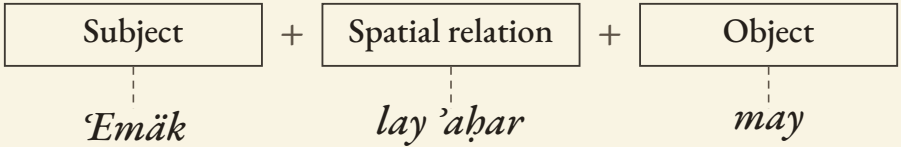
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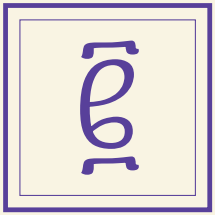
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# The five Ge’ez games

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*Section 2*

# Imitation

# Game 2: objective

- After observing sufficiently many examples of Ge'ez sentences you are now capable of producing not only syntactically valid, but also semantically meaningful sentences! Give it a try.

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- For example:
  - "Carlos is at ETH" is **semantically valid!**

# Game 2: objective

- After observing sufficiently many examples of Ge'ez sentences you are now capable of producing not only syntactically valid, but also semantically meaningful sentences! Give it a try.
- For example:
  - "Carlos is at ETH" is **semantically valid!**
  - "ETH is at Carlos" is **not semantically valid!**

# Insight

We don’t need to understand a language in order to produce meaningful sentences!

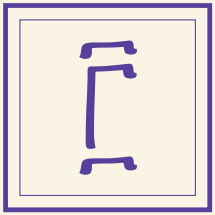
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# The five Ge’ez games

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*Section 3*

# Prediction

# Prediction

| | | |
|---|---|---|
|  | ዕድል<br>Probability | Language models can be described as **probabilistic trees**. |
|  | ሰንጠረዥ<br>Table | **Probabilistic trees** can be represented with **tables**. |
|  | ሥልጠና<br>Learning | **Tables** can be learned from language texts. |
|  | የውይይት ሮቦት<br>Chatbot | The **table** can be used as a **chatbot**. |

# Game 3: Prediction

Alice plays tennis

# Game 3: Prediction

Alice plays tennis

Alice plays golf

# Game 3: Prediction

Alice plays tennis

Alice plays golf

Alice sings

# Game 3: Prediction

Alice plays tennis

Alice plays golf

Alice sings

Bob plays tennis

# Game 3: Prediction

Alice plays tennis

Alice plays golf

Alice sings

Bob plays tennis

Bob plays chess

# Probability trees

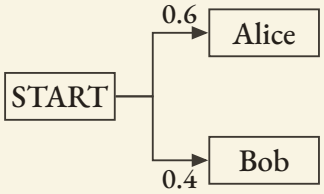
| Alice plays tennis | Alice plays golf | Alice sings | Bob plays tennis | Bob plays chess |
|---|---|---|---|---|

START

*We can organize all these sentences into a probability tree!*

# Probability trees

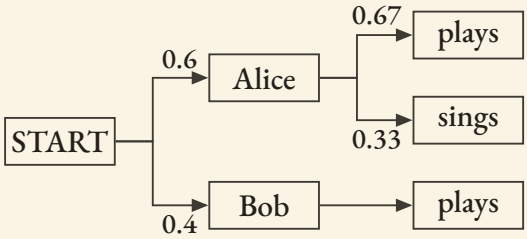
| Alice plays tennis | Alice plays golf | Alice sings | Bob plays tennis | Bob plays chess |
|---|---|---|---|---|




*We can organize all these sentences into a probability tree!*

# Probability trees

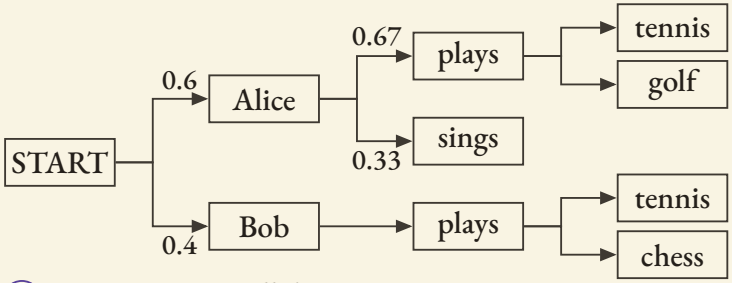
| Alice plays tennis | Alice plays golf | Alice sings | Bob plays tennis | Bob plays chess |
|---|---|---|---|---|




*We can organize all these sentences into a probability tree!*

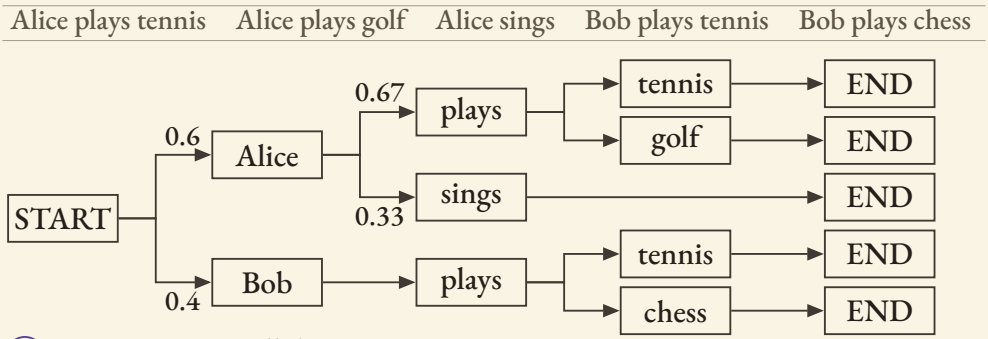
# Probability trees

| Alice plays tennis | Alice plays golf | Alice sings | Bob plays tennis | Bob plays chess |
|---|---|---|---|---|




*We can organize all these sentences into a probability tree!*

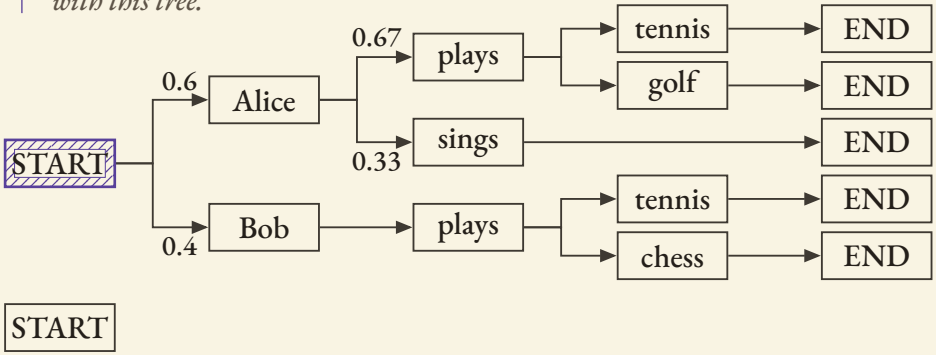
# Probability trees




*We can organize all these sentences into a probability tree!*

# Text generation as sampling

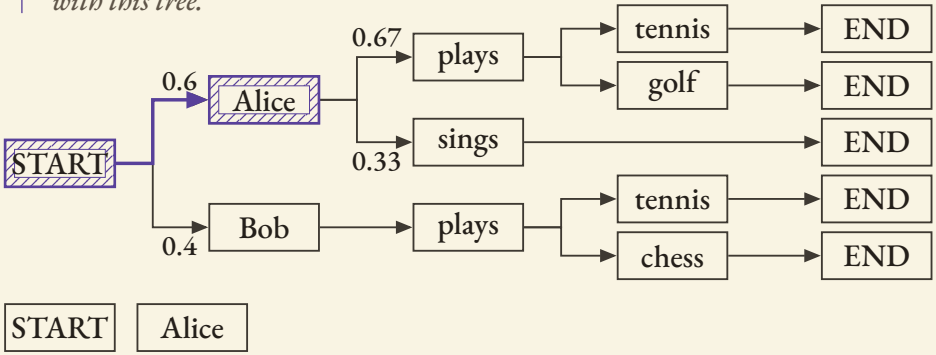
*We can now produce sentences with this tree.*




START

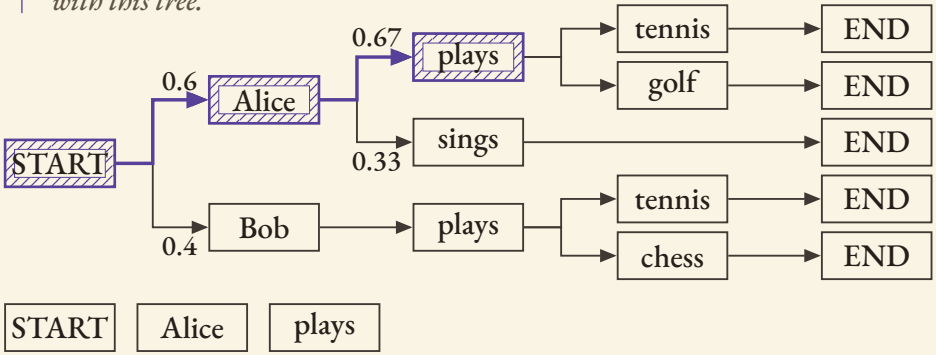
# Text generation as sampling

*We can now produce sentences with this tree.*



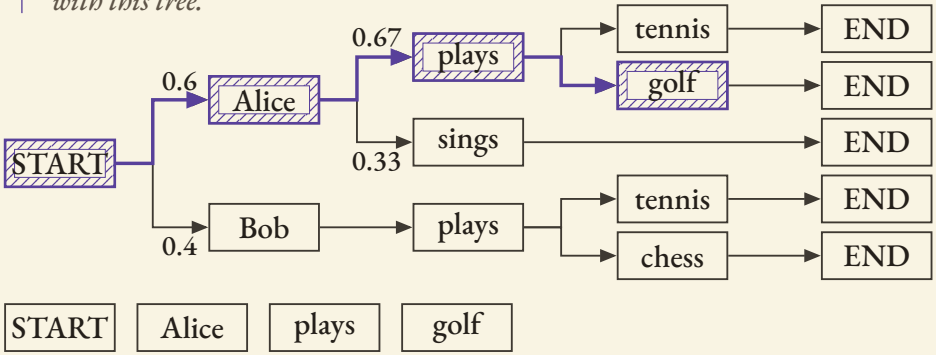
# Text generation as sampling

*We can now produce sentences with this tree.*



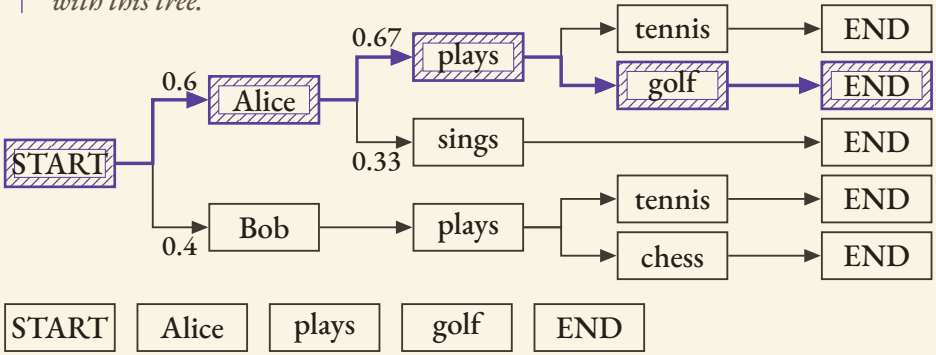
# Text generation as sampling

*We can now produce sentences with this tree.*



# Text generation as sampling

*We can now produce sentences with this tree.*



# Insight: We can represent a language model with a table!

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | | | | | | | | |
| Alice | | | | | | | | |
| Bob | | | | | | | | |
| Alice plays | | | | | | | | |
| Alice sings | | | | | | | | |
| Bob plays | | | | | | | | |
| Alice plays golf | | | | | | | | |
| Alice plays tennis | | | | | | | | |
| Bob plays chess | | | | | | | | |
| Bob plays tennis | | | | | | | | |



*We omit the keyword "START" in some prefixes for the sake of conciseness.*

# Insight: We can represent a language model with a table!

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | | | | | | | | |
| Alice | | | | | | | | |
| Bob | | | | | | | | |
| Alice plays | | | | | | | | |
| Alice sings | | | | | | | | |
| Bob plays | | | | | | | | |
| Alice plays golf | | | | | | | | |
| Alice plays tennis | | | | | | | | |
| Bob plays chess | | | | | | | | |
| Bob plays tennis | | | | | | | | |

How likely is sings after Alice?

# Insight: We can represent a language model with a table!

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | | | | | | | | |
| Alice | | | | | | | | |
| Bob | | | | | | | | |
| Alice plays | | | | | | | | |
| Alice sings | | | | | | | | |
| Bob plays | | | | | | | | |
| Alice plays golf | | | | | | | | |
| Alice plays tennis | | | | | | | | |
| Bob plays chess | | | | | | | | |
| Bob plays tennis | | | | | | | | |

How likely is golf after Bob plays?

# Insight: We can represent a language model with a table!

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | | | | | | | | |
| Alice | | | | | | | | |
| Bob | | | | | | | | |
| Alice plays | | | | | | | | |
| Alice sings | | | | | | | | |
| Bob plays | | | | | | | | |
| Alice plays golf | | | | | | | | |
| Alice plays tennis | | | | | | | | |
| Bob plays chess | | | | | | | | |
| Bob plays tennis | | | | | | | | |

How likely is plays after Bob plays tennis?

# Insight: We can represent a language model with a table!

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |
| Alice | | | 0.67 | 0.33 | | | | |
| Bob | | | 1.0 | | | | | |
| Alice plays | | | | | | 0.5 | 0.5 | |
| Alice sings | | | | | | | | 1.0 |
| Bob plays | | | | | 0.5 | | 0.5 | |
| Alice plays golf | | | | | | | | 1.0 |
| Alice plays tennis | | | | | | | | 1.0 |
| Bob plays chess | | | | | | | | 1.0 |
| Bob plays tennis | | | | | | | | 1.0 |



*A blank cell is a chance of 0: no sentence of the corpus goes that way.*

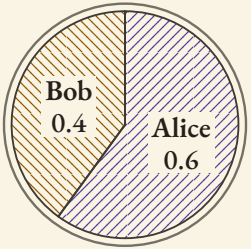
# Sampling: spin a roulette

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |

# Sampling: spin a roulette

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |

*Each slice is as big as the chance of its word. A word with no chance gets no slice.*

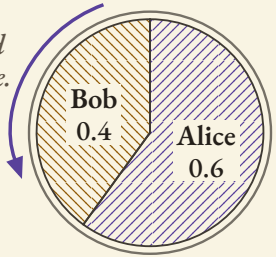


# Sampling: spin a roulette

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |

*Each slice is as big as the chance of its word. A word with no chance gets no slice.*

*Spin the roulette.*

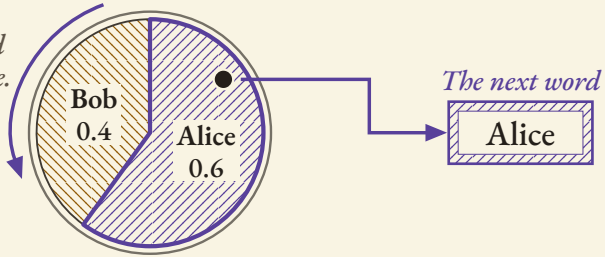


# Sampling: spin a roulette

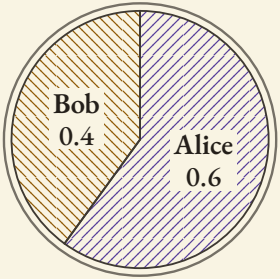
| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |

*Each slice is as big as the chance of its word. A word with no chance gets no slice.*

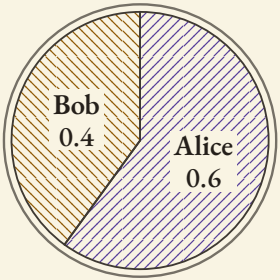
*Spin the roulette.*



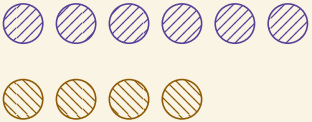
# A bigger slice comes out more often



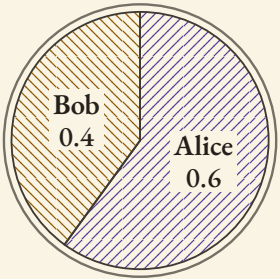
# A bigger slice comes out more often




Ten spins, on average



# A bigger slice comes out more often




Ten spins, on average

Alice

Bob

*A real run of ten spins can differ a little. The more spins, the closer the shares get to 6 in 10 and 4 in 10.*

# We can now generate sentences by iterative sampling.

START

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |
| Alice | | | 0.67 | 0.33 | | | | |
| Bob | | | 1.0 | | | | | |
| Alice plays | | | | | | 0.5 | 0.5 | |
| Alice sings | | | | | | | | 1.0 |
| Bob plays | | | | | 0.5 | | 0.5 | |
| Alice plays golf | | | | | | | | 1.0 |
| Alice plays tennis | | | | | | | | 1.0 |
| Bob plays chess | | | | | | | | 1.0 |
| Bob plays tennis | | | | | | | | 1.0 |

# We can now generate sentences by iterative sampling.

START Alice

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |
| Alice | | | 0.67 | 0.33 | | | | |
| Bob | | | 1.0 | | | | | |
| Alice plays | | | | | | 0.5 | 0.5 | |
| Alice sings | | | | | | | | 1.0 |
| Bob plays | | | | | 0.5 | | 0.5 | |
| Alice plays golf | | | | | | | | 1.0 |
| Alice plays tennis | | | | | | | | 1.0 |
| Bob plays chess | | | | | | | | 1.0 |
| Bob plays tennis | | | | | | | | 1.0 |

# We can now generate sentences by iterative sampling.

START Alice plays

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |
| Alice | | | 0.67 | 0.33 | | | | |
| Bob | | | 1.0 | | | | | |
| Alice plays | | | | | | 0.5 | 0.5 | |
| Alice sings | | | | | | | | 1.0 |
| Bob plays | | | | | 0.5 | | 0.5 | |
| Alice plays golf | | | | | | | | 1.0 |
| Alice plays tennis | | | | | | | | 1.0 |
| Bob plays chess | | | | | | | | 1.0 |
| Bob plays tennis | | | | | | | | 1.0 |

# We can now generate sentences by iterative sampling.

START Alice plays golf

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |
| Alice | | | 0.67 | 0.33 | | | | |
| Bob | | | 1.0 | | | | | |
| Alice plays | | | | | | 0.5 | 0.5 | |
| Alice sings | | | | | | | | 1.0 |
| Bob plays | | | | | 0.5 | | 0.5 | |
| Alice plays golf | | | | | | | | 1.0 |
| Alice plays tennis | | | | | | | | 1.0 |
| Bob plays chess | | | | | | | | 1.0 |
| Bob plays tennis | | | | | | | | 1.0 |

# We can now generate sentences by iterative sampling.

START Alice plays golf END

| Prefix | Alice | Bob | plays | sings | chess | golf | tennis | END |
|---|---|---|---|---|---|---|---|---|
| START | 0.6 | 0.4 | | | | | | |
| Alice | | | 0.67 | 0.33 | | | | |
| Bob | | | 1.0 | | | | | |
| Alice plays | | | | | | 0.5 | 0.5 | |
| Alice sings | | | | | | | | 1.0 |
| Bob plays | | | | | 0.5 | | 0.5 | |
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| Bob plays chess | | | | | | | | 1.0 |
| Bob plays tennis | | | | | | | | 1.0 |

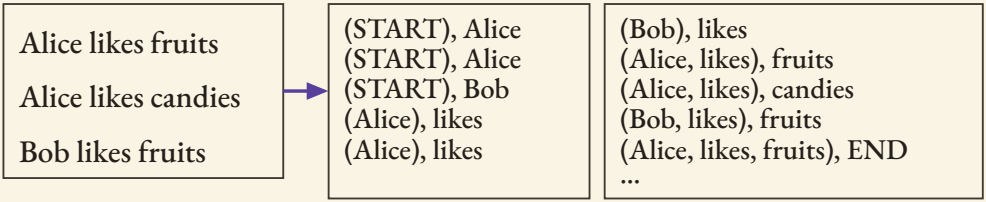
# Training an autoregressive language model

Alice likes fruits

Alice likes candies

Bob likes fruits

# Training an autoregressive language model



# Training an autoregressive language model

Alice likes fruits

Alice likes candies

Bob likes fruits

(START), Alice
(START), Alice
(START), Bob
(Alice), likes
(Alice), likes

(Bob), likes
(Alice, likes), fruits
(Alice, likes), candies
(Bob, likes), fruits
(Alice, likes, fruits), END
...

| Prefix | Alice | Bob | likes | candies | fruits | END |
|---|---|---|---|---|---|---|
| START | 2/3 | 1/3 | | | | |
| Alice | | | 1 | | | |
| Bob | | | 1 | | | |
| Alice likes | | | | 1/2 | 1/2 | |
| Bob likes | | | | | 1 | |
| ... | | | | | | |

# Your first chatbot!

What does Alice like? Alice plays tennis

What does Alice like? Alice plays golf

What does Alice like? Alice sings

What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess

# Your first chatbot!

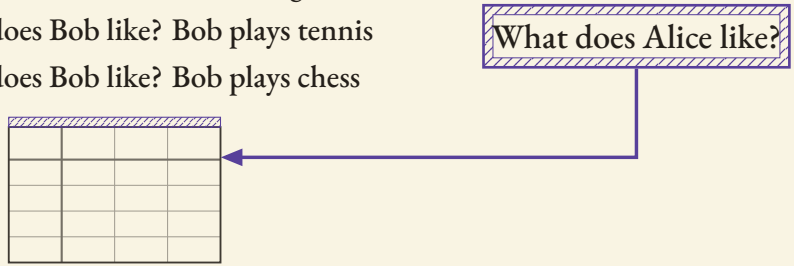
What does Alice like? Alice plays tennis

What does Alice like? Alice plays golf

What does Alice like? Alice sings

What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess



# Your first chatbot!

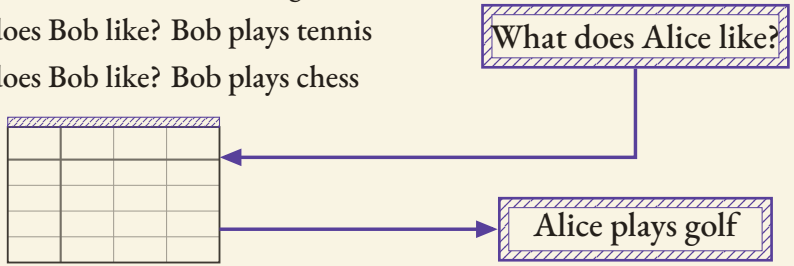
What does Alice like? Alice plays tennis

What does Alice like? Alice plays golf

What does Alice like? Alice sings

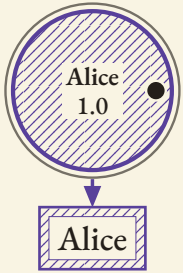
What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess



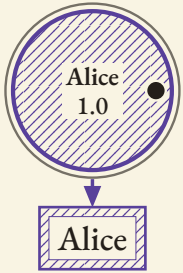
# A chatbot answers one word at a time

What does Alice like?

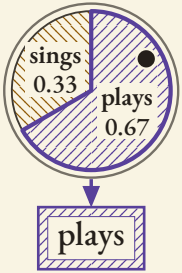


# A chatbot answers one word at a time

What does Alice like?

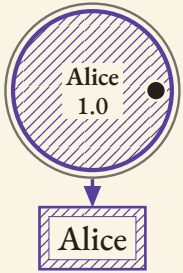



What does Alice like? Alice

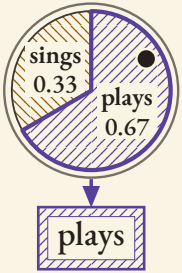


# A chatbot answers one word at a time

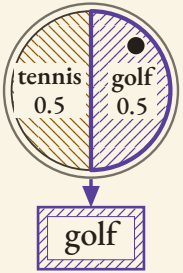
What does Alice like?



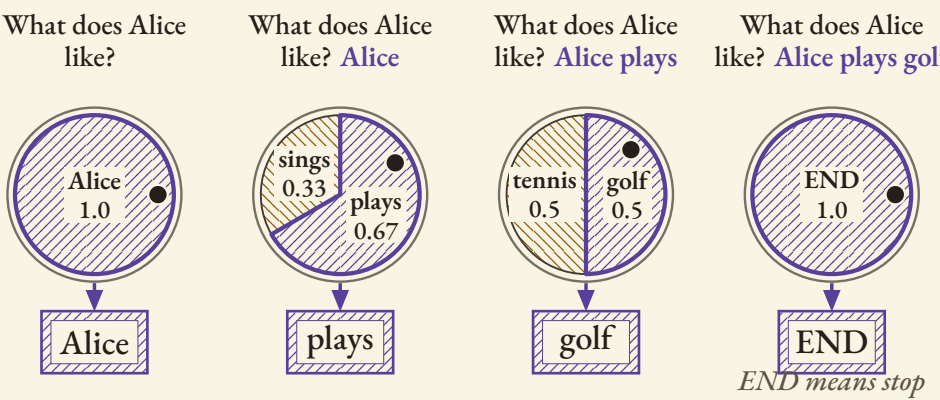

What does Alice like? Alice



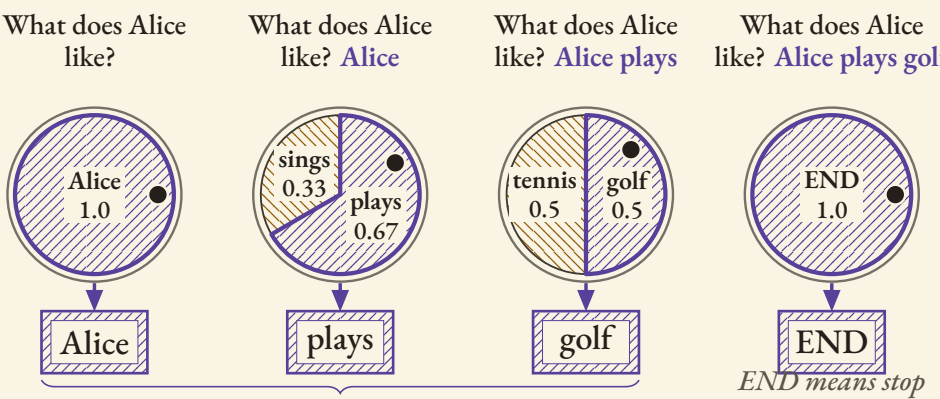

What does Alice like? Alice plays



# A chatbot answers one word at a time



# A chatbot answers one word at a time



# Summary

- Language models can be described as probabilistic trees.

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- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.

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- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.
- In our toy, a **table** can be learned from a set of training sentences, by generating all combinations of prefix and continuation, and then calculating probabilities in the corresponding table.

# Summary

- Language models can be described as probabilistic trees.
- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.
- In our toy, a **table** can be learned from a set of training sentences, by generating all combinations of prefix and continuation, and then calculating probabilities in the corresponding table.
- In our toy, a conversation can then be carried out, by just taking the user's prompt, and using the **table** to generate **probable continuations**.

# The five Ge’ez games

| | | |
|---|---|---|
|  | ሥርዓት<br>**Structure** | We can extract structure from unfamiliar text without knowing a single word. |
|  | ምሳል<br>**Imitation** | By spotting patterns, we can compose new sentences in a language we don’t understand. |
|  | ትንቢት<br>**Prediction** | A language model picks the next word by probabilities conditioned on what came before. |
|  | ምልክት<br>**Representation** | A prefix becomes an embedding, and the embedding gives the chances of the next word. |
|  | ቅንብር<br>**Transformer** | An LLM computes the chances, draws a word, appends it, and repeats. |